SHRINE uncertainties: step-by-step derivation of the structure-parameter and DM errors

Setup and notation

S23 is Sutinjo et al. (2023, ApJ 954, 37); "S23 Eq. N" uses the published numbering.

- At one trial DM d the dedispersed intensity is $\boldsymbol{i}$ (N samples). Its true parts are $\boldsymbol{i} = \boldsymbol{s} + \boldsymbol{n}$: $\boldsymbol{s}$ is the noiseless burst and $\boldsymbol{n}$ is noise with $\langle \boldsymbol{n} \rangle = 0$ and $\mathrm{Cov}(\boldsymbol{n}) = \langle \boldsymbol{n}\boldsymbol{n}^T \rangle$.
- $\langle \cdot \rangle$ is the average over noise realisations: the same burst at the same DM, with new noise.
- $\boldsymbol{S}$ is the smoothing matrix (symmetric) and $\boldsymbol{D}_1$ the $(N-1) \times N$ first difference, $(\boldsymbol{D}_1 \boldsymbol{x})_j = x_{j+1} - x_j$ (S23 Eq. 2).
- $\boldsymbol{S} = \boldsymbol{C}\,\mathrm{diag}[f(k)]\,\boldsymbol{C}^T$ (S23 Eq. 15), with $\boldsymbol{C}$ the orthonormal DCT-II matrix and $f(k)$ S23's Butterworth response (Eq. 16) with cutoff k_c and order O. eig_k (S23 Eq. 14) are the eigenvalues of $\boldsymbol{D}_1^T \boldsymbol{D}_1$. SHRINE's `get_kc` places k_c where the burst's DCT spectrum falls to the noise level.
- The structure parameter is $\mathcal{S} = \|\boldsymbol{D}_1 \boldsymbol{S} \boldsymbol{i}\|$, the norm of S23 Eq. 3.
- S23 Eq. 4 splits the data as $\boldsymbol{i} = \tilde{\boldsymbol{i}} + \boldsymbol{\Delta}_{\boldsymbol{i}}$ with $\tilde{\boldsymbol{i}} = \boldsymbol{S}\boldsymbol{i}$ and $\boldsymbol{\Delta}_{\boldsymbol{i}} = \boldsymbol{i} - \boldsymbol{S}\boldsymbol{i}$. These are *estimates* of signal and noise built with $\boldsymbol{S}$. Substituting the true parts $\boldsymbol{i} = \boldsymbol{s} + \boldsymbol{n}$,

$$\tilde{\boldsymbol{i}} = \boldsymbol{S}\boldsymbol{s} + \boldsymbol{S}\boldsymbol{n}, \qquad \boldsymbol{\Delta}_{\boldsymbol{i}} = (\boldsymbol{s} - \boldsymbol{S}\boldsymbol{s}) + (\boldsymbol{n} - \boldsymbol{S}\boldsymbol{n}). \tag{1}$$

  Neither part is pure. $\tilde{\boldsymbol{i}}$ contains the in-band noise $\boldsymbol{S}\boldsymbol{n}$ that smoothing cannot remove. $\boldsymbol{\Delta}_{\boldsymbol{i}}$ contains the noise that smoothing removed, $\boldsymbol{n} - \boldsymbol{S}\boldsymbol{n}$, and also the burst structure that smoothing removed, $\boldsymbol{s} - \boldsymbol{S}\boldsymbol{s}$: the *leakage* term. The leakage is the same in every realisation, so it is not noise, but S23 treat all of $\boldsymbol{\Delta}_{\boldsymbol{i}}$ as noise.

Step 1: split the structure parameter by the true parts

Define

$$\boldsymbol{a} = \boldsymbol{D}_1 \boldsymbol{S} \boldsymbol{s}, \qquad \boldsymbol{e} = \boldsymbol{D}_1 \boldsymbol{S} \boldsymbol{n}, \qquad \hat{\boldsymbol{a}} = \boldsymbol{a}/\|\boldsymbol{a}\|. \tag{2}$$

Then $\boldsymbol{D}_1 \boldsymbol{S} \boldsymbol{i} = \boldsymbol{a} + \boldsymbol{e}$ exactly, so $\mathcal{S} = \|\boldsymbol{a} + \boldsymbol{e}\|$. $\boldsymbol{a}$ is fixed (it depends only on the burst); $\boldsymbol{e}$ is random with zero mean.

$\boldsymbol{s}$ and $\boldsymbol{n}$ are not known in practice. The split is still what tells us which statistic is the correct error; Step 4 replaces the unknowns by estimates.

Step 2: only the noise along $\boldsymbol{a}$ matters at first order

Split the noise into a part along $\hat{\boldsymbol{a}}$ and a part perpendicular to it:

$$\boldsymbol{e} = e_{\parallel}\,\hat{\boldsymbol{a}} + \boldsymbol{e}_{\perp}, \qquad e_{\parallel} = \hat{\boldsymbol{a}}^T \boldsymbol{e}. \tag{3}$$

$e_\parallel$ is a signed scalar. Because $\boldsymbol{e}_\perp \perp \hat{\boldsymbol{a}}$, Pythagoras gives

$$\|\boldsymbol{a}+\boldsymbol{e}\|^2 = (\|\boldsymbol{a}\|+e_\parallel)^2 + \|\boldsymbol{e}_\perp\|^2. \tag{4}$$

For $\|\boldsymbol{e}_\perp\| \ll \|\boldsymbol{a}\|$, $\sqrt{X^2+Y^2} \simeq X + Y^2/2X$. Replacing $\|\boldsymbol{a}\|+e_\parallel$ by $\|\boldsymbol{a}\|$ in the last term changes it only at third order, so to second order in the noise

$$\boxed{\mathcal{S} \simeq \|\boldsymbol{a}\| + e_\parallel + \frac{\|\boldsymbol{e}_\perp\|^2}{2\|\boldsymbol{a}\|}} \tag{5}$$

- $e_\parallel$ is the only noise term at first order. It sets the 1σ scatter of $\mathcal{S}$.
- $\|\boldsymbol{e}_\perp\|^2/2\|\boldsymbol{a}\| \geq 0$ is a second-order positive bias, not an uncertainty.
- S23's bounds (Eq. 8) are triangle inequalities. They are reached only when the added vector is parallel to the structure vector (here $\boldsymbol{e}_\perp = 0$, all of the noise along $\hat{\boldsymbol{a}}$). They are bounds, not 1σ errors.

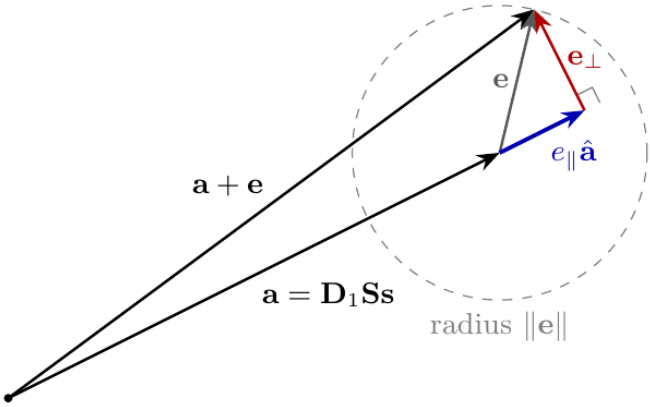


Figure 1: *Visualisation only (not a derivation step).* The noise $\boldsymbol{e}$ added to $\boldsymbol{a}$, split into $e_\parallel\hat{\boldsymbol{a}}$ and $\boldsymbol{e}_\perp$. Only $e_\parallel$ changes the length at first order; $\boldsymbol{e}_\perp$ tilts the vector. The dashed circle has radius $\|\boldsymbol{e}\|$.

Step 3: variance of $e_\parallel$

3a. Weights. $e_\parallel = \hat{\boldsymbol{a}}^T \boldsymbol{D}_1 \boldsymbol{S}\boldsymbol{n}$. Define the row of weights

$$\boldsymbol{w} = \hat{\boldsymbol{a}}^T \boldsymbol{D}_1 \boldsymbol{S} \qquad (1\times N), \qquad \text{so} \qquad e_\parallel = \boldsymbol{w}\boldsymbol{n} = \sum_t w_t n_t. \tag{6}$$

w_t is the first-order change of $\mathcal{S}$ per unit change of noise sample n_t. It is fixed by the burst.

3b. Mean. Each w_t is a constant and each n_t is one value of the vector $\boldsymbol{n}$. If $\langle n_t\rangle = 0$, then $\langle e_\parallel\rangle = \sum_t w_t\langle n_t\rangle = 0$. (A constant offset in $\boldsymbol{n}$ is removed by $\boldsymbol{D}_1$ anyway.)

3c. Variance. With zero mean, $\mathrm{Var}(X) = \langle X^2\rangle - \langle X\rangle^2 = \langle X^2\rangle$, so $\sigma^2 = \langle e_\parallel^2\rangle$. Write the square as the sum times itself, counting the second copy with its own dummy label t' so that every term of one copy multiplies every term of the other:

$$\sigma^2 = \Big\langle \Big(\sum_t w_t n_t\Big)\Big(\sum_{t'} w_{t'} n_{t'}\Big)\Big\rangle = \sum_t\sum_{t'} w_t w_{t'} \langle n_t n_{t'}\rangle = \boldsymbol{w}\ \mathrm{Cov}(\boldsymbol{n})\ \boldsymbol{w}^T. \tag{7}$$

$\langle n_t n_{t'}\rangle$ is element (t,t') of $\mathrm{Cov}(\boldsymbol{n})$. Its diagonal ($t'=t$) is the variance σ_t^2.

3d. Noise uncorrelated between samples. $\langle n_t n_{t'}\rangle = \sigma_t^2$ if $t' = t$ and 0 otherwise, so the t' sum keeps one term, $\sum_{t'} w_{t'}\langle n_t n_{t'}\rangle = w_t\sigma_t^2$, and

$$\boxed{\sigma^2 = \sum_t w_t^2 \sigma_t^2} \tag{8}$$

For white noise, $\sigma^2 = \sigma_n^2 \sum_t w_t^2$. σ is the 1σ uncertainty of $\mathcal{S}$, and the relative uncertainty is $\sigma/\mathcal{S}$.

Step 4: evaluating σ from the data

4a. Direction. $\hat{\boldsymbol{a}}$ is unknown. Estimate it from the data,

$$\hat{\boldsymbol{a}} \simeq \frac{\boldsymbol{D}_1\boldsymbol{S}\boldsymbol{i}}{\|\boldsymbol{D}_1\boldsymbol{S}\boldsymbol{i}\|} = \frac{\boldsymbol{a}+\boldsymbol{e}}{\|\boldsymbol{a}+\boldsymbol{e}\|}. \tag{9}$$

This is tilted from the true $\hat{\boldsymbol{a}}$ by an angle θ with $\tan\theta = \|\boldsymbol{e}_\perp\|/(\|\boldsymbol{a}\| + e_\parallel)$, so $\theta \simeq \|\boldsymbol{e}_\perp\|/\|\boldsymbol{a}\|$.

4b. Size of $\|\boldsymbol{e}_\perp\|$. For any fixed matrix $\boldsymbol{M}$ and noise uncorrelated between samples,

$$\|\boldsymbol{M}\boldsymbol{n}\|^2 = \sum_j \Big(\sum_t M_{jt} n_t\Big)^2 = \sum_j \sum_t \sum_{t'} M_{jt} M_{jt'} n_t n_{t'} \quad \Rightarrow \quad \langle\|\boldsymbol{M}\boldsymbol{n}\|^2\rangle = \sum_j \sum_t M_{jt}^2 \sigma_t^2, \tag{10}$$

by the same collapse of the t' sum as in 3d. With $\boldsymbol{M} = \boldsymbol{D}_1\boldsymbol{S}$ this is $\langle\|\boldsymbol{e}\|^2\rangle$. Define

$$N_e = \langle\|\boldsymbol{e}\|^2\rangle/\sigma^2 : \tag{11}$$

the total noise power in $\boldsymbol{e}$ divided by the part along $\hat{\boldsymbol{a}}$ ($\langle e_\parallel^2\rangle = \sigma^2$), i.e. how many directions' worth of noise $\boldsymbol{e}$ carries. It would be $N-1$ (the length of $\boldsymbol{e}$) for isotropic noise; in our simulations ($N = 2048$, $k_c \simeq 40$) $N_e \simeq 40$–45. Since $\langle\|\boldsymbol{e}_\perp\|^2\rangle = (N_e - 1)\sigma^2$,

$$\theta \simeq \sqrt{N_e - 1}\,\sigma/\|\boldsymbol{a}\| \simeq 6\,\sigma/\|\boldsymbol{a}\|. \tag{12}$$

The estimated direction is good when $\|\boldsymbol{a}\| \gg 6\sigma$. The relative error it introduces in σ is of order θ. In our simulations $\theta \simeq 0.2$ at burst amplitude 0.5, and σ computed from the data matched the Monte Carlo to within $\sim$10%.

4c. Weights. $\boldsymbol{w} = \hat{\boldsymbol{a}}^T\boldsymbol{D}_1\boldsymbol{S}$. First $(\hat{\boldsymbol{a}}^T\boldsymbol{D}_1)_t = \hat{a}_{t-1} - \hat{a}_t$ (with $\hat{a}_0 = \hat{a}_N = 0$; $\hat{\boldsymbol{a}}$ has $N-1$ entries), then smooth the result with $\boldsymbol{S}$.

4d. Noise level. For white noise with standard deviation σ_n per sample, take σ_n from off-pulse data, or from the on-pulse DCT coefficients above $k = N/4$, where the burst has no power: $\hat{\sigma}_n = 1.4826\,\mathrm{median}_{k>N/4}|(\boldsymbol{C}^T\boldsymbol{i})_k|$ (the orthonormal DCT leaves the white-noise variance unchanged in every mode). For self-noise, take σ_t^2 as the local mean square of the series rebuilt from the modes $k > N/4$ only, divided by 3/4 (the fraction of white-noise variance that reconstruction keeps). Then use Eq. (8).

Step 5: what S23's radius measures (S23 Eqs. 6, 7, 19)

5a. Split S23's residual. Apply $\boldsymbol{D}_1\boldsymbol{S}$ to $\boldsymbol{\Delta_i}$ from Eq. (1):

$$\boldsymbol{D}_1\boldsymbol{S}\boldsymbol{\Delta_i} = \boldsymbol{d}_s + \boldsymbol{d}_n, \qquad \boldsymbol{d}_s = \boldsymbol{D}_1\boldsymbol{S}(\boldsymbol{s} - \boldsymbol{S}\boldsymbol{s}), \qquad \boldsymbol{d}_n = \boldsymbol{D}_1\boldsymbol{S}(\boldsymbol{n} - \boldsymbol{S}\boldsymbol{n}). \tag{13}$$

$\boldsymbol{d}_s$ is the leakage term after $\boldsymbol{D}_1\boldsymbol{S}$: burst structure that smoothing removed, smoothed again. It is the same in every realisation, so it is not noise. $\boldsymbol{d}_n$ is random with zero mean.

5b. Expected squared radius. The cross term averages to zero, $2\boldsymbol{d}_s^T\langle\boldsymbol{d}_n\rangle = 0$, so

$$\langle\|\boldsymbol{D}_1\boldsymbol{S}\boldsymbol{\Delta}_{\boldsymbol{i}}\|^2\rangle = \|\boldsymbol{d}_s\|^2 + \langle\|\boldsymbol{d}_n\|^2\rangle. \tag{14}$$

$\boldsymbol{d}_n = (\boldsymbol{D}_1\boldsymbol{S} - \boldsymbol{D}_1\boldsymbol{S}\boldsymbol{S})\,\boldsymbol{n}$ is a fixed matrix times $\boldsymbol{n}$, so $\langle\|\boldsymbol{d}_n\|^2\rangle$ follows from Eq. (10) with $\boldsymbol{M} = \boldsymbol{D}_1\boldsymbol{S} - \boldsymbol{D}_1\boldsymbol{S}\boldsymbol{S}$. It is not zero: $\langle\boldsymbol{d}_n\rangle = 0$, but $\|\boldsymbol{d}_n\|^2$ is a sum of squares.

5c. Ratio to the true variance. Divide by σ^2:

$$\boxed{\frac{\langle\|\boldsymbol{D}_1\boldsymbol{S}\boldsymbol{\Delta}_{\boldsymbol{i}}\|^2\rangle}{\sigma^2} = \frac{\langle\|\boldsymbol{d}_n\|^2\rangle}{\sigma^2}\left(1 + \frac{\|\boldsymbol{d}_s\|^2}{\langle\|\boldsymbol{d}_n\|^2\rangle}\right)} \tag{15}$$

- *First factor: noise in all directions instead of along $\hat{\boldsymbol{a}}$.* It equals $N_e\,\langle\|\boldsymbol{d}_n\|^2\rangle/\langle\|\boldsymbol{e}\|^2\rangle$. $\boldsymbol{e} = \boldsymbol{D}_1\boldsymbol{S}\boldsymbol{n}$ is the derivative of the smoothed noise. $\boldsymbol{d}_n = \boldsymbol{D}_1\boldsymbol{S}(\boldsymbol{n} - \boldsymbol{S}\boldsymbol{n})$ is the derivative of the smoothed version of the noise that smoothing removed, and smoothing it again removes most of it. Per DCT mode k (S23 Eq. 17), $\boldsymbol{D}_1\boldsymbol{S}$ multiplies by $\mathrm{eig}_k^{1/2} f(k)$ and $\boldsymbol{D}_1\boldsymbol{S} - \boldsymbol{D}_1\boldsymbol{S}\boldsymbol{S}$ by $\mathrm{eig}_k^{1/2} f(k)[1 - f(k)]$. Since $f(1-f) \le 1/4$, with the maximum at k_c where the filter passes half ($f = \frac{1}{2}$), only the noise near k_c survives. It would vanish for a brick-wall filter. In our simulations $\langle\|\boldsymbol{d}_n\|^2\rangle \simeq \langle\|\boldsymbol{e}\|^2\rangle/7.7$, so the first factor is (40–45)/7.7 ≃ 5–6.
- *Second factor: burst leakage.* $\boldsymbol{d}_s$ lies near k_c, where `get_kc` places the cutoff at the point where the burst falls to the noise level, so $\|\boldsymbol{d}_s\|^2 \gtrsim \langle\|\boldsymbol{d}_n\|^2\rangle$. $\boldsymbol{d}_s$ scales with burst amplitude, so $\|\boldsymbol{d}_s\|^2/\langle\|\boldsymbol{d}_n\|^2\rangle \propto (\mathrm{S/N})^2$ at fixed k_c. In our simulations the second factor is 4.1 and 12.8 for burst amplitudes 0.5 and 1.0 ($k_c = 38$, 40).

5d. S23's per-sample reading. $\|\boldsymbol{D}_1\boldsymbol{S}\boldsymbol{\Delta}_{\boldsymbol{i}}\|/\sqrt{N-1}$ (text after S23 Eq. 6) assumes the residual is white over $N-1$ samples. It is concentrated near k_c, so this *under*estimates σ, by a factor 5–10 in our simulations.

5e. Normalisation (S23 Eq. 7). A relative uncertainty is the uncertainty of a quantity divided by that same quantity. σ (Step 3) is the scatter of $\mathcal{S} = \|\boldsymbol{D}_1\boldsymbol{S}\boldsymbol{i}\|$, the structure parameter that SHRINE reports and maximises over DM, and $\mathcal{S} \simeq \|\boldsymbol{a}\| + e_{\|}$ (Step 2). So the relative uncertainty is $\sigma/\mathcal{S} \simeq \sigma/\|\boldsymbol{a}\|$.

S23 divide instead by $\|\boldsymbol{D}_1\boldsymbol{S}\tilde{\boldsymbol{i}}\| = \|\boldsymbol{D}_1\boldsymbol{S}\boldsymbol{S}\boldsymbol{i}\|$ (in the code, `_uncertainty_calc` divides by the norm of `combined*fL*c`). This comes from S23 Eq. 5, which treats $\boldsymbol{D}_1\boldsymbol{S}\tilde{\boldsymbol{i}}$ as the signal part of $\boldsymbol{D}_1\tilde{\boldsymbol{i}}$. But $\boldsymbol{D}_1\boldsymbol{S}\tilde{\boldsymbol{i}}$ is the data smoothed twice: its burst part is $\boldsymbol{D}_1\boldsymbol{S}\boldsymbol{S}\boldsymbol{s}$, not $\boldsymbol{a} = \boldsymbol{D}_1\boldsymbol{S}\boldsymbol{s}$. Per DCT mode k (S23 Eq. 17), with c_k the DCT coefficients of $\boldsymbol{s}$,

$$\|\boldsymbol{D}_1\boldsymbol{S}\boldsymbol{s}\|^2 = \sum_k \mathrm{eig}_k\, f(k)^2 c_k^2, \qquad \|\boldsymbol{D}_1\boldsymbol{S}\boldsymbol{S}\boldsymbol{s}\|^2 = \sum_k \mathrm{eig}_k\, f(k)^4 c_k^2. \tag{16}$$

Since $f(k) \le 1$, every mode is reduced, so $\|\boldsymbol{D}_1\boldsymbol{S}\boldsymbol{S}\boldsymbol{s}\| < \|\boldsymbol{a}\|$. The loss is largest near k_c, where $f \simeq \frac{1}{2}$ and where the band pass $\mathrm{eig}_k^{1/2} f(k)$ of the structure parameter peaks (S23 Fig. 2). In our simulations the double-smoothed norm is 5–6% below $\|\boldsymbol{a}\|$ at any S/N. It is a fixed bias set by the burst shape and k_c, not noise, so it does not shrink as the S/N rises, and it inflates the relative uncertainty by the same 5–6%.

5f. Root cause. The leakage ($\boldsymbol{d}_s$) and the normalisation both come from treating $\tilde{\boldsymbol{i}}$ as the true signal and $\boldsymbol{\Delta}_{\boldsymbol{i}}$ as the noise (S23 Eq. 4), when both mix burst and noise (Eq. 1). The first factor of Eq. (15) is the separate radius-versus-projection problem of Step 2.

Step 6: DM uncertainty

6a. Structure curve. At each trial DM d, Steps 1–2 give

$$\mathcal{S}(d) \simeq \|\boldsymbol{a}(d)\| + e_{\|}(d), \tag{17}$$

with $\boldsymbol{a}(d) = \boldsymbol{D}_1 \boldsymbol{S} \boldsymbol{s}(d)$ and $e_{\|}(d)$ the noise along $\hat{\boldsymbol{a}}(d)$. The second-order bias varies slowly with d and is dropped. Let d_0 be the peak of the noise-free curve $\|\boldsymbol{a}(d)\|$. For a burst with the same profile in every channel this is the true DM. With drift it can differ; that systematic offset is separate and not covered here.

6b. Expand about d_0.

$$\|\boldsymbol{a}(d)\| \simeq \|\boldsymbol{a}(d_0)\| + \tfrac{1}{2}\|\boldsymbol{a}\|''(d_0)\,(d-d_0)^2, \tag{18}$$

$$e_{\|}(d) \simeq e_{\|}(d_0) + e'_{\|}(d_0)\,(d-d_0). \tag{19}$$

The linear term of the first line is absent because $\|\boldsymbol{a}\|'(d_0) = 0$ at the peak. $\|\boldsymbol{a}\|''(d_0) < 0$ is the curvature, the rate at which the slope changes; near its peak any smooth curve is close to a parabola. So

$$\mathcal{S}(d) \simeq \|\boldsymbol{a}(d_0)\| + e_{\|}(d_0) + e'_{\|}(d_0)\,(d-d_0) + \tfrac{1}{2}\|\boldsymbol{a}\|''(d_0)\,(d-d_0)^2. \tag{20}$$

The noise value $e_{\|}(d_0)$ lifts the whole curve and does not move the peak. The noise slope $e'_{\|}(d_0)$ tilts it: $\mathcal{S}'(d_0) = e'_{\|}(d_0) \neq 0$, so the curve is no longer flat at d_0 and the peak moves.

6c. Peak position. At the measured peak $\hat{d}$, $\mathcal{S}'(\hat{d}) = 0$:

$$e'_{\|}(d_0) + \|\boldsymbol{a}\|''(d_0)\,(\hat{d}-d_0) = 0 \quad \Rightarrow \quad \hat{d} - d_0 = -\frac{e'_{\|}(d_0)}{\|\boldsymbol{a}\|''(d_0)}. \tag{21}$$

$\|\boldsymbol{a}\|''(d_0)$ is fixed and $\mathrm{std}(X/K) = \mathrm{std}(X)/|K|$, so

$$\sigma_{\mathrm{DM}} = \frac{\mathrm{std}[e'_{\|}(d_0)]}{|\|\boldsymbol{a}\|''(d_0)|}. \tag{22}$$

6d. $e_{\|}(d)$ in terms of the channel noise. From Step 3, $e_{\|}(d) = \sum_t w(t;d)\,n(t;d)$, where $w(t;d)$ is the weight row computed from the burst dedispersed at d (it changes with d because the dedispersed burst does; it does not depend on the burst amplitude). Dedispersion gives

$$n(t;d) = \sum_{\nu} x_\nu(t + d\,\tau_\nu), \qquad \tau_\nu = K_{\mathrm{DM}}\,(\nu^{-2} - \nu_{\mathrm{ref}}^{-2})/t_{\mathrm{samp}}. \tag{23}$$

- t is the time-sample index; $x_\nu(t)$ is the raw noise in channel ν at sample t.
- $K_{\mathrm{DM}} \simeq 4.149\,\mathrm{ms\,GHz^2\,(pc\,cm^{-3})^{-1}}$, so τ_ν is in samples per $\mathrm{pc\,cm^{-3}}$ and $d\,\tau_\nu$ (DM times τ_ν) is channel ν's delay in samples; dedispersion advances the channel by it. Fractional shifts are applied as FFT phase shifts.
- ν_{ref} is a free choice (see 6g).

Hence

$$e_{\|}(d) = \sum_{\nu}\sum_{t} w(t;d)\,x_\nu(t + d\,\tau_\nu). \tag{24}$$

6e. Move the shift onto the weights. Relabel $t \to t - d\,\tau_\nu$ in each channel:

$$e_\parallel(d) = \sum_\nu \sum_t w(t - d\,\tau_\nu; d)\, x_\nu(t). \tag{25}$$

The pairs are unchanged; only the counting label moves. With a one-sample shift, $\sum_t w_t x_{t+1} = \sum_t w_{t-1} x_t = \ldots + w_4 x_5 + w_5 x_6 + w_6 x_7 + \ldots$: the shift moves to the other factor with the opposite sign. This is exact for circular (FFT) shifts, and otherwise holds up to edge samples where the weights are $\simeq 0$.

Why: $x_\nu(t)$ no longer depends on d. All the d-dependence is in the weights, which are smooth (they have passed through $\boldsymbol{S}$) and can be differentiated. White noise cannot.

6f. Differentiate. In each term $x_\nu(t)$ is a constant factor, so

$$e'_\parallel(d_0) = \sum_\nu \sum_t w'_\nu(t)\, x_\nu(t), \qquad w'_\nu(t) = \frac{\partial}{\partial d}\, w(t - d\,\tau_\nu; d)\Big|_{d_0}. \tag{26}$$

$w'_\nu(t)$ is the change, per unit DM, of the weight that multiplies raw noise sample $x_\nu(t)$. It changes because the channel slides under the weight row (by τ_ν samples per unit DM, so differently in each channel) and because the row itself changes shape as the dedispersed burst changes.

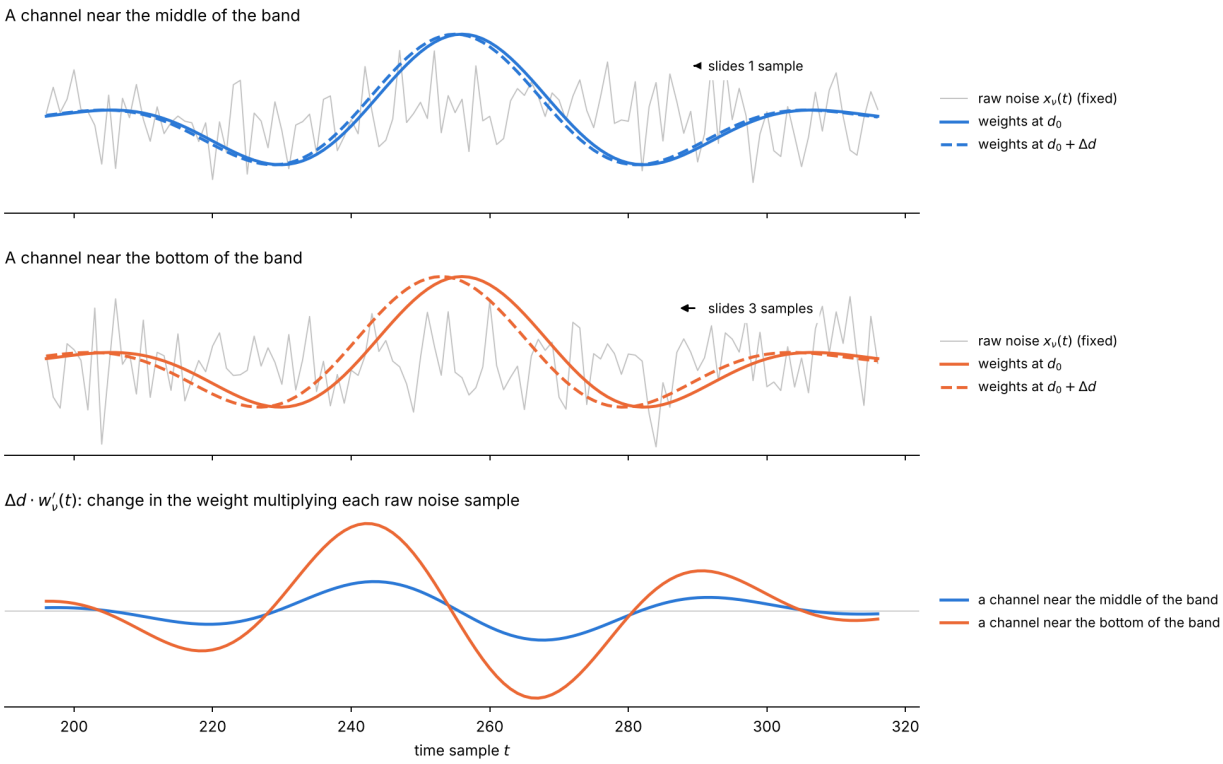


Figure 2: *Visualisation only (not a derivation step).* The same weight row over the fixed raw noise of two channels, at d_0 (solid) and $d_0 + \Delta d$ (dashed). The two channels slide different distances under the row; the bottom panel is the resulting change of the weight multiplying each raw noise sample, $\Delta d \cdot w'_\nu(t)$. Drawn for a burst with the same profile in every channel, so the row only slides; with drift or scattering it also changes shape. Delays are measured from the amplitude-weighted mean delay.

6g. Variance. Within one noise realisation, $x_\nu(t)$ is one dynamic spectrum of noise; ν and t are summation labels, $e_\parallel(d)$ is one curve against d, and $e'_\parallel(d_0)$ is one number. Across realisations,

the burst, the weights, w'_ν, τ_ν and d_0 stay fixed and $x_\nu(t)$ changes. Assume $x_\nu(t)$ has zero mean, is independent between channels and between samples, and has variance σ_ν^2. Then, exactly as in Step 3 with the label (ν, t) in place of t: $\langle e'_\parallel(d_0)\rangle = 0$, so Var $= \langle e'_\parallel(d_0)^2\rangle$, and only the terms with $\nu' = \nu$, $t' = t$ survive:

$$\mathrm{Var}[e'_\parallel(d_0)] = \sum_\nu \sigma_\nu^2 \sum_t w'_\nu(t)^2, \tag{27}$$

$$\boxed{\sigma_{\mathrm{DM}} = \frac{\left(\sum_\nu \sigma_\nu^2 \sum_t w'_\nu(t)^2\right)^{1/2}}{|\|\boldsymbol{a}\|''(d_0)|}} \tag{28}$$

- $\langle e'_\parallel(d_0)\rangle = 0$: to first order $\hat{d}$ scatters symmetrically about d_0.
- This holds for any burst morphology (drift, scattering, profile evolution).
- For self-noise, use $\sigma_\nu(t)^2$ inside the t sum.
- It does not depend on ν_{ref}: changing ν_{ref} shifts every channel and the weight row by the same amount, so $w(t - d\,\tau_\nu; d)$ is unchanged.

6h. Evaluating σ_{DM} from the data. Everything at d_0 is evaluated at the measured peak $\hat{d}$ (as $\hat{\boldsymbol{a}}$ was in Step 4). None of the following assumes the same profile in every channel.

1. *Curvature $\|\boldsymbol{a}\|''$.* The measured curve is $\mathcal{S}(d) = \|\boldsymbol{a}(d)\| + e_\parallel(d) + \ldots$, so its curvature is the one we want plus noise curvature. Two estimates:
   - *Fit:* least-squares parabola $\mathcal{S}(d) \simeq c_0 + c_1 d + c_2 d^2$ over the trial DMs with $|d - \hat{d}| \le W$; then $\|\boldsymbol{a}\|'' \simeq 2c_2$ (and $\hat{d} = -c_1/2c_2$).
   - *Finite difference:* $\|\boldsymbol{a}\|'' \simeq [\mathcal{S}(\hat{d}+h) - 2\mathcal{S}(\hat{d}) + \mathcal{S}(\hat{d}-h)]/h^2$, dedispersing at $\hat{d} \pm h$ if they are not on the grid.

   $e_\parallel(d)$ is smooth in d, so a narrow window or small step also picks up the curvature of the noise, while too wide a window leaves the region where $\|\boldsymbol{a}(d)\|$ is parabolic. Use $W \simeq 2$–$3\,\sigma_{\mathrm{DM}}$ or $h \simeq 2\,\sigma_{\mathrm{DM}}$: start from a guess, compute σ_{DM}, and refit. In the drifting-burst test (below) both were unbiased to $\simeq 1\%$, with 16–84% scatter of $\pm$12–15% at S/N 14 and $\pm$5–7% at S/N 38; $W = 1.5\,\sigma_{\mathrm{DM}}$ or $h = \sigma_{\mathrm{DM}}$ increased the scatter by up to a factor of 2. Check that the result is stable when W or h is changed.

2. *Weight rows.* For each needed d, dedisperse the data, set $\hat{\boldsymbol{a}}(d) \simeq \boldsymbol{D}_1\boldsymbol{S}\boldsymbol{i}(d)/\|\boldsymbol{D}_1\boldsymbol{S}\boldsymbol{i}(d)\|$ and $w(t; d) = \hat{\boldsymbol{a}}(d)^T \boldsymbol{D}_1 \boldsymbol{S}$ (Step 4c).

3. $w'_\nu(t)$ by central difference (the recipe for computing it, not a derivation step):

$$w'_\nu(t) \simeq \frac{w(t - (\hat{d}+\delta)\tau_\nu;\, \hat{d}+\delta) - w(t - (\hat{d}-\delta)\tau_\nu;\, \hat{d}-\delta)}{2\delta}. \tag{29}$$

   That is: compute the row at $\hat{d} \pm \delta$, read each at the position where channel ν's raw sample t lands at that DM, and subtract. Expanding $g(\hat{d} \pm \delta) = g \pm \delta g' + \frac{1}{2}\delta^2 g'' \pm \ldots$ shows the error is $\propto \delta^2$ (one-sided: $\propto \delta$). Take $\delta \simeq \sigma_{\mathrm{DM}}/2$; in the test, $\delta = 0.1$ and $0.5\,\sigma_{\mathrm{DM}}$ agreed at S/N 38, while at S/N 14 the noise in the data-built rows raised the numerator of Eq. (28) by 10% ($\delta = 0.5\,\sigma_{\mathrm{DM}}$) and 13% ($\delta = 0.1\,\sigma_{\mathrm{DM}}$).

4. σ_ν from off-pulse data in each channel (or from each channel's $k > N/4$ DCT band, as in 4d).

5. *Alternative without derivatives:* pass $J \sim 100$–200 off-pulse noise segments through the same dedispersion, add each noise curve $e_\parallel^{(j)}(d) = \sum_t w(t; d)\, n^{(j)}(t; d)$ to the structure curve, and take the scatter of the peak over j (`struct_max_errors.dm_uncertainty`).

6i. Special case (optional: only for the closed form and its scaling; not needed above). Suppose channel ν has profile $A_\nu\, p(t)$ with the same unit-amplitude p in every channel, and let $\bar\tau = \sum_\nu A_\nu\tau_\nu / \sum_\nu A_\nu$ be the amplitude-weighted mean delay. At $d = d_0 + \Delta d$ channel ν is misaligned by $\Delta d\,\tau_\nu$, so the summed burst is $\sum_\nu A_\nu\, p(t + \Delta d\,\tau_\nu)$. Split each shift as $\Delta d\,\tau_\nu = \Delta d\,\bar\tau + \Delta d\,(\tau_\nu - \bar\tau)$. The common part slides the whole series rigidly. Expanding each channel in the remaining shift,

$$\sum_\nu A_\nu\, p\big(t + \Delta d(\tau_\nu - \bar\tau)\big) \simeq \Big(\sum_\nu A_\nu\Big) p(t) + \Delta d \sum_\nu A_\nu(\tau_\nu - \bar\tau)\,\dot p(t) + \tfrac{1}{2}\Delta d^2 \sum_\nu A_\nu(\tau_\nu - \bar\tau)^2\,\ddot p(t), \quad (30)$$

and the middle term is zero by the definition of $\bar\tau$. (With two equal channels at $\tau_\nu - \bar\tau = \pm 1$: $p(t+\Delta d) + p(t-\Delta d) \simeq 2p(t) + \Delta d^2 \ddot p(t)$; the linear terms cancel and the sum only broadens.) So at first order the summed burst, and with it the weight row, only slides by $\Delta d\,\bar\tau$: $w(t; d) \simeq w(t + \Delta d\,\bar\tau; d_0)$. Then $w(t - d\,\tau_\nu; d) \simeq w(t - d\,\tau_\nu + \Delta d\,\bar\tau; d_0)$ depends on d only through its position, and the chain rule gives

$$w'_\nu(t) = -(\tau_\nu - \bar\tau)\,\dot w(t - d_0\tau_\nu; d_0), \quad (31)$$

with $\dot w$ the time-slope of the weight row. The sum of $\dot w^2$ over all samples does not depend on the shift, so $\mathrm{Var}[e'_\parallel(d_0)] = \sum_\nu \sigma_\nu^2(\tau_\nu - \bar\tau)^2 \sum_t \dot w(t; d_0)^2$.

For the curvature, the rigid slide does not change $\|\boldsymbol{a}\|$. Applying the linear $\boldsymbol{D}_1\boldsymbol{S}$ to the expansion above (which has the common slide removed), $\boldsymbol{a}(d) \simeq \boldsymbol{a}(d_0) + \frac{1}{2}\Delta d^2 \big[\sum_\nu A_\nu(\tau_\nu - \bar\tau)^2\big] \boldsymbol{D}_1\boldsymbol{S}\ddot{\boldsymbol{p}}$. By Step 2 the length changes by $\hat{\boldsymbol{a}}^T$ times this (the perpendicular part enters at Δd^4), and $\hat{\boldsymbol{a}}^T\boldsymbol{D}_1\boldsymbol{S} = \boldsymbol{w}$, so

$$\|\boldsymbol{a}\|''(d_0) = \Big[\sum_\nu A_\nu(\tau_\nu - \bar\tau)^2\Big] \sum_t w(t; d_0)\,\ddot p(t), \quad (32)$$

$$\sigma_{\mathrm{DM}} = \frac{\big(\sum_\nu \sigma_\nu^2(\tau_\nu - \bar\tau)^2\big)^{1/2} \big(\sum_t \dot w(t; d_0)^2\big)^{1/2}}{\big[\sum_\nu A_\nu(\tau_\nu - \bar\tau)^2\big]\,\big|\sum_t w(t; d_0)\,\ddot p(t)\big|}. \quad (33)$$

With equal noise σ_x and amplitude A in all N_ν channels this is

$$\sigma_{\mathrm{DM}} = \frac{\sigma_x}{A}\,\frac{1}{\big[\sum_\nu(\tau_\nu - \bar\tau)^2\big]^{1/2}}\,\frac{\big(\sum_t \dot w^2\big)^{1/2}}{|\sum_t w\ddot p|}. \quad (34)$$

Here $(\sigma_x/A)/\sqrt{N_\nu}$ is 1/(S/N of the summed burst), $[\sum_\nu(\tau_\nu - \bar\tau)^2/N_\nu]^{1/2}$ is the rms delay spread, and $(\sum_t \dot w^2)^{1/2}/|\sum_t w\ddot p|$ is a time in samples set by the burst structure. So $\sigma_{\mathrm{DM}} \simeq$ (burst timescale)/(S/N $\times$ rms delay spread). This fails for drift, scattering or profile evolution; use Eq. (28).

Step 7: S23's DM interval (S23 Eq. 11, Sects. 3.2, 4)

S23 keep the trial DMs whose structure is within the noise of the peak, $\mathcal{S}(\hat d) - \mathcal{S}(d) \le \kappa\,\sigma_{\hat d, d}$, with $\kappa = 1$ and the radius $\|\boldsymbol{D}_1\boldsymbol{S}(\boldsymbol{\Delta}_{\boldsymbol{i},d} - \boldsymbol{\Delta}_{\boldsymbol{i},\hat d})\|$ as $\sigma_{\hat d, d}$. Near the peak, $\mathcal{S}(\hat d) - \mathcal{S}(d) \simeq \frac{1}{2}|\|\boldsymbol{a}\|''|(d - \hat d)^2$, and the noise of the difference is $e_\parallel(\hat d) - e_\parallel(d) \simeq e'_\parallel\,(\hat d - d)$, with standard deviation $\mathrm{std}[e'_\parallel]\,|d - \hat d|$. The edge of the interval is where the two are equal:

$$|d - \hat d| = \frac{2\kappa\ \mathrm{std}[e'_\parallel]}{|\|\boldsymbol{a}\|''|} = 2\kappa\,\sigma_{\mathrm{DM}}. \quad (35)$$

$\kappa = \frac{1}{2}$ gives $\pm 1\sigma$; $\kappa = 1$ gives $\pm 2\sigma_{\mathrm{DM}}$ even with the correct noise. With the radius (Step 5) as well, the S23 half-width was 8.7 and 13 times the Monte-Carlo scatter of $\hat d$ in our simulations, growing with S/N. Quoting Eq. (28) directly avoids the interval altogether.

Table 1: Steps 2–5 and 7: `sim.py` (two-component burst, 84 channels, $N = 2048$, white noise; 200 realisations at the true DM, 150 over the DM grid for the last two rows). The Monte-Carlo standard deviations are uncertain by $\simeq 5\%$.

burst amplitude (k_c)	0.5 (38)	1.0 (40)
Monte-Carlo std($\mathcal{S}$)	0.240	0.241
Eq. (8), true burst	0.252	0.256
bias $\langle\mathcal{S}\rangle - \|\boldsymbol{a}\|$: $(\langle\|\boldsymbol{e}\|^2\rangle - \sigma^2)/2\|\boldsymbol{a}\|$ / Monte Carlo	0.180 / 0.172	0.102 / 0.099
S23 radius $\|\boldsymbol{D}_1\boldsymbol{S}\boldsymbol{\Delta}_{\boldsymbol{i}}\|$ (median)	1.167	2.205
S23 radius / Monte Carlo	4.9	9.2
Eq. (15): first factor, second factor	5.2, 4.1	5.8, 12.8
$\sqrt{\text{Eq. (15)}}$	4.6	8.6
S23 per-sample reading $\|\boldsymbol{D}_1\boldsymbol{S}\boldsymbol{\Delta}_{\boldsymbol{i}}\|/\sqrt{N-1}$	0.026	0.049
Monte-Carlo std($\hat{d}$) [pc cm^{-3}]	0.00355	0.00160
S23 half-width / Monte Carlo	8.7	13

Numerical checks

Step 6. Toy model: 32 channels over 1.1–1.5 GHz, $N = 1024$, channel amplitudes $\propto \nu^{-1.5}$, $\nu_{\rm ref}$ at the top channel, Butterworth $k_c = 40$, $O = 5$; Monte Carlo of the full DM search (300–1000 realisations). S/N is the peak of the summed burst over the standard deviation of the summed noise per sample. Ratios are formula/Monte Carlo.

- *Drifting two-component burst* (the components drift by 24 and 36 samples across the band, so the same-profile model does not hold). Eq. (28) with the true burst: 0.97 ± 0.03 against the scatter of the first-order curve $\|\boldsymbol{a}(d)\| + e_\parallel(d)$; against the full search 0.91 ± 0.03 at summed S/N 14 and 0.96 ± 0.03 at S/N 38. The shortfall is second order and falls with S/N. Estimated from each realisation's data alone (6h, $W = 3\,\sigma_{\rm DM}$, $\delta = 0.5\,\sigma_{\rm DM}$): median 0.98 (16–84%: 0.89–1.12) at S/N 14 and 0.94 (0.90–0.99) at S/N 38; at S/N 14 the noise in the data-built rows (6h) partly offsets the second-order shortfall.
- *Same profile in every channel* (6i). The closed-form $\|\boldsymbol{a}\|''(d_0)$ matches the numerical curvature to four significant figures; the closed-form $\sigma_{\rm DM}$ gives 0.96 ± 0.03 at S/N 11 and 0.99 ± 0.02 at S/N 35.

Scripts: `checks/general_check.py`, `checks/linear_check.py`, `checks/same_profile_check.py`.

Changes to S23

1. Eq. 4: add the true split $\boldsymbol{i} = \boldsymbol{s} + \boldsymbol{n}$; $\boldsymbol{\Delta}_{\boldsymbol{i}}$ contains burst structure and $\tilde{\boldsymbol{i}}$ contains in-band noise.
2. Eqs. 5, 8 and the text after them: replace the bounds by Eq. (5).
3. Eqs. 6, 19: replace the radius by $\sigma^2 = \sum_t w_t^2\sigma_t^2$, $\boldsymbol{w} = \hat{\boldsymbol{a}}^T\boldsymbol{D}_1\boldsymbol{S}$; drop the $\sqrt{N-1}$ reading.
4. Eq. 7: divide by $\mathcal{S} = \|\boldsymbol{D}_1\boldsymbol{S}\boldsymbol{i}\|$, not $\|\boldsymbol{D}_1\boldsymbol{S}\tilde{\boldsymbol{i}}\|$.
5. Eqs. 9–11 and the DM interval: quote $\sigma_{\rm DM}$ from Eq. (28), with $\|\boldsymbol{a}\|''$ from a fit or finite difference (6h); if an interval test is kept, use $\kappa = \frac{1}{2}$ and the noise std$[e'_\parallel]\,|d - \hat{d}|$.